Portfolio 4: NoSQL and Advanced Data Models

Project: Stretto Technologies Database System

NoSQL Model Selected: Document Store (MongoDB)

1. Explanation of Suitability

While the relational PostgreSQL schema designed in Portfolio 1 effectively enforces strict financial integrity, a **Document Store NoSQL model** is exceptionally suitable for the inventory and order-retrieval domains of Stretto Technologies. A simple Key-Value store lacks the depth to easily query individual line items, and a Graph database is over-engineered for basic sales transactions.

Electronics sourcing involves highly diverse product specifications. For example, a smartphone requires attributes for battery capacity and IMEI numbers, while a gaming console requires attributes for controller count and digital/disc edition flags. In a rigid relational database, accommodating these varying attributes results in sparse tables filled with NULL values. A NoSQL Document Store resolves this by utilizing a flexible, schema-less structure, allowing Stretto Technologies to absorb dynamic product specifications without altering the database schema.

Furthermore, this model improves read performance through denormalization. In the relational model, retrieving a complete customer receipt required joining four distinct tables (invoices, customers, invoice_items, and items). In the Document model, all related data is embedded into a single, comprehensive JSON/BSON document, allowing the application to retrieve the entire transaction in a single database read operation.

2. Document Model Example

To demonstrate this denormalized approach, the core entities of a Stretto Technologies transaction have been aggregated into a single nested document. The customer details and the array of sourced items are embedded directly inside the parent invoice document.

JSON


```
{
  "invoice_number": "INV_NOSQL_001",
  "create_date": "2026-10-22T10:30:00Z",
  "total_amount": 1550.00,
  "customer_details": {
    "customer_id": 1,
```

```
    "full_name": "Eric Gbagi",
    "email": "eric.gbagi@example.com"
  },
  "purchased_items": [
    {
      "item_id": 5,
      "item_name": "iPhone 15 Pro",
      "supplier_name": "OA Plug Device",
      "quantity": 1,
      "unit_price": 1000.00,
      "device_identifiers": "IMEI-847593028"
    },
    {
      "item_id": 12,
      "item_name": "PlayStation 5",
      "supplier_name": "Gaming Hub NG",
      "quantity": 1,
      "unit_price": 500.00,
      "device_identifiers": "SN-9948302A"
    }
  ]
}
```


3. Implementation Evidence

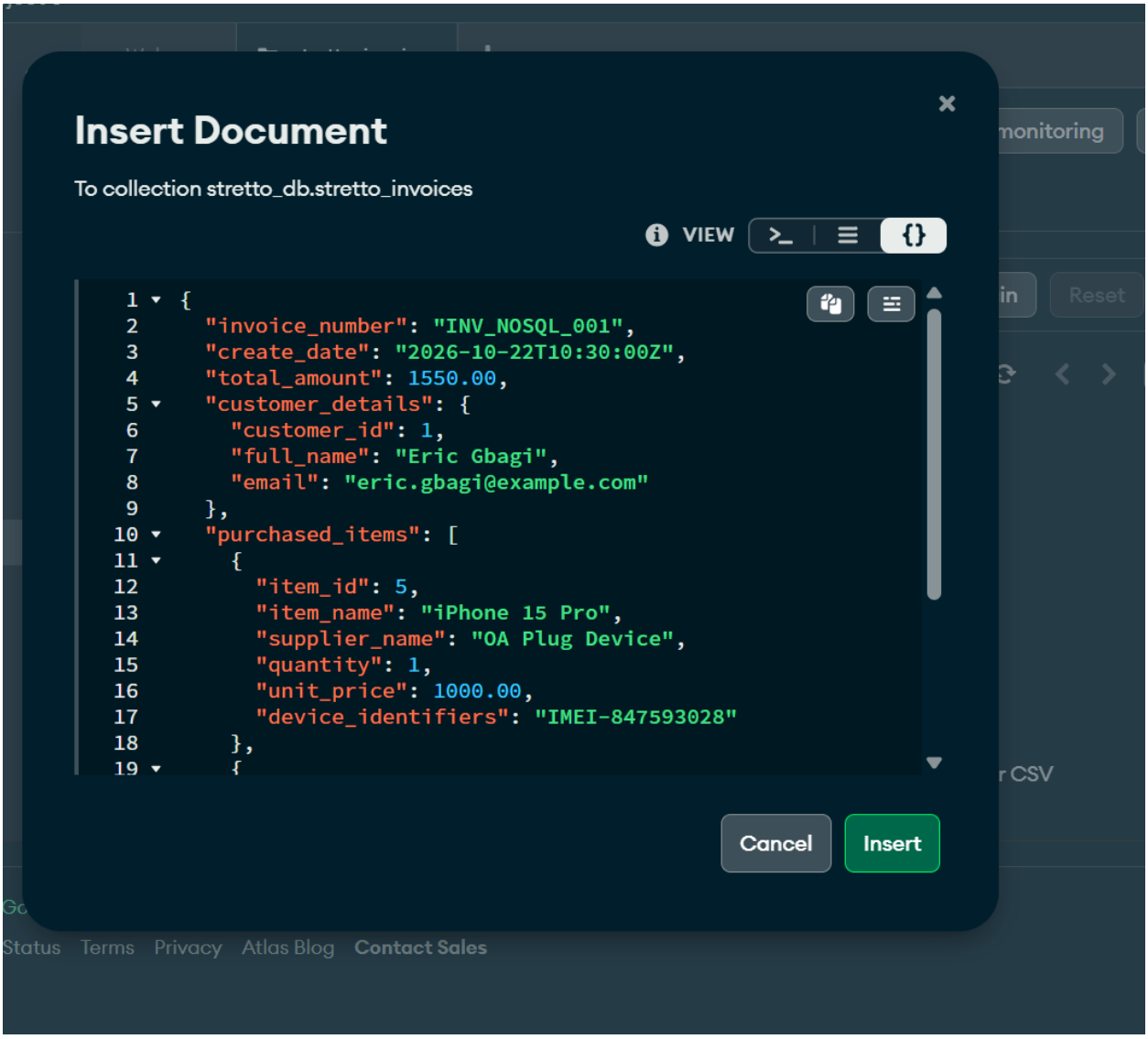
The conceptual document model was physically implemented using the MongoDB Atlas cloud database interface to provide evidence of operational viability.

A. **Document Insertion** The complex, nested invoice document was successfully inserted into the stretto_invoices collection using the Atlas data explorer. The following JSON payload was utilized:

JSON


```
{
 "invoice_number": "INV_NOSQL_001",
 "create_date": "2026-10-22T10:30:00Z",
 "total_amount": 1550.00,
 "customer_details": {
  "customer_id": 1,
  "full_name": "Eric Gbagi",
  "email": "eric.gbagi@example.com"
 },
 "purchased_items": [
  {
   "item_id": 5,
   "item_name": "iPhone 15 Pro",
   "supplier_name": "OA Plug Device",
   "quantity": 1,
   "unit_price": 1000.00,
   "device_identifiers": "IMEI-847593028"
  },
  {
   "item_id": 12,
   "item_name": "PlayStation 5",
   "supplier_name": "Gaming Hub NG",
   "quantity": 1,
```

```
    "unit_price": 500.00,
    "device_identifiers": "SN-9948302A"
  }
 ]
}
```



B. Data Retrieval (Querying the Nested Document) To verify the extraction capabilities of the Document Store, the database was queried using a specific filter and projection. The query isolates the specific invoice and extracts only the invoice number, the embedded customer name, and the embedded item array, suppressing the default MongoDB ObjectID.

Filter Applied:

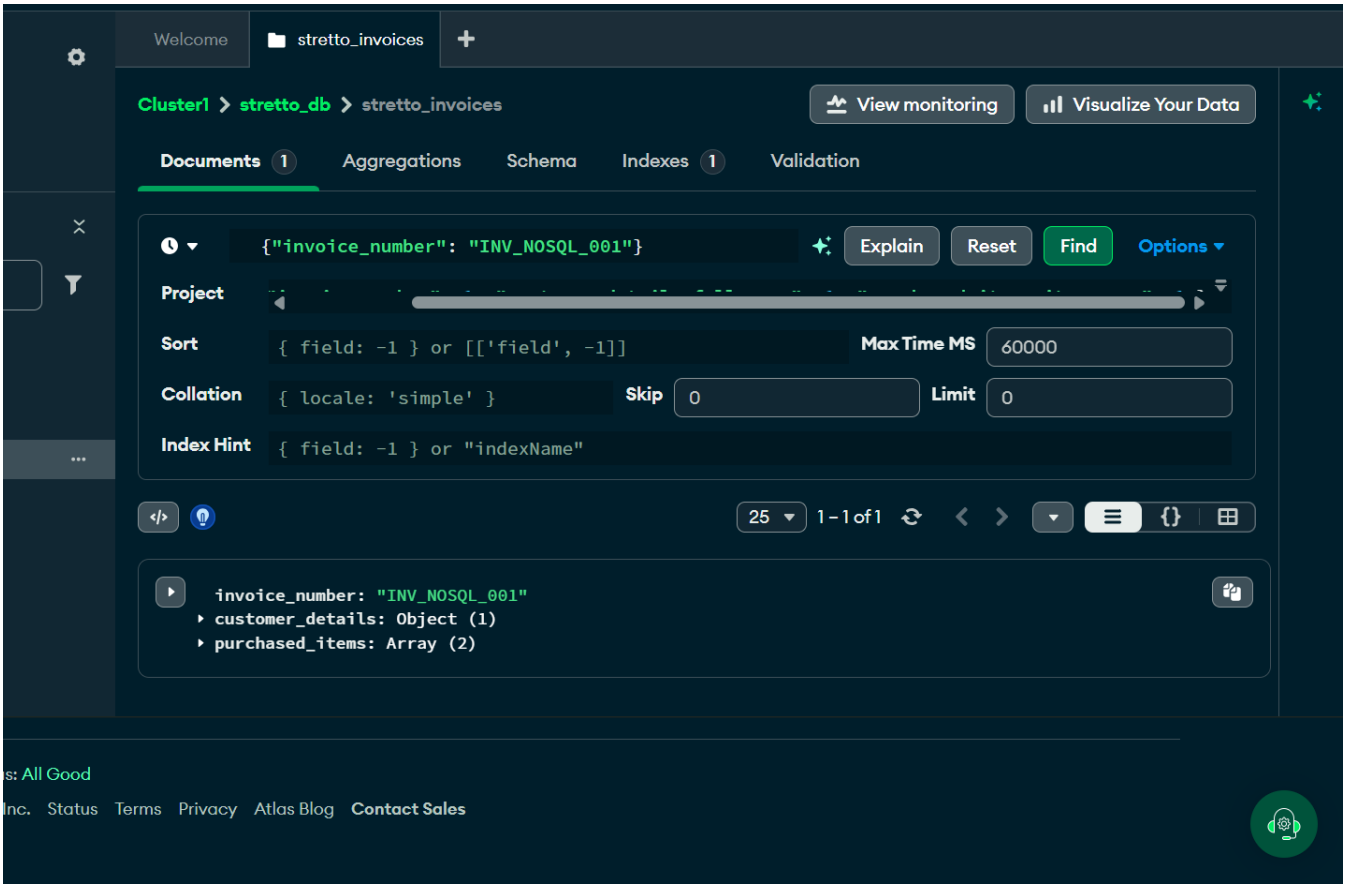
JSON

```
{ "invoice_number": "INV_NOSQL_001" }
```

Projection Applied:

JSON

```
{
  "_id": 0,
  "invoice_number": 1,
  "customer_details.full_name": 1,
  "purchased_items.item_name": 1
}
```


Welcome
stretto_invoices
Cluster1 > stretto_db > stretto_invoices
View monitoring
Visualize Your Data
Documents 1
Aggregations
Schema
Indexes 1
Validation
{"invoice_number": "INV_NOSQL_001"}
Explain
Reset
Find
Options
Project
Sort
{ field: -1 } or [['field', -1]]
Max Time MS
60000
Collation
{ locale: 'simple' }
Skip
0
Limit
0
Index Hint
{ field: -1 } or "indexName"
25
1 – 1 of 1
invoice_number: "INV_NOSQL_001"
customer_details: Object (1)
purchased_items: Array (2)
s: All Good
Inc.
Status
Terms
Privacy
Atlas Blog
Contact Sales